

A SPECIAL EDITION OF

MERIDIAN

The Home Cockpit — an ultra deep dive

 A flight simulator cockpit fitted with a Garmin G1000 glass panel

Source: Wikimedia Commons — a Garmin G1000-equipped flight simulator cockpit. Representative image, honestly labelled.

THE HARDWARE

From a US\$50 stick to a five-figure simpit: the buyer's ladder the community actually climbs, tier by tier

VATSIM

190,000 people fly and control for nobody, for free, on a volunteer network older than the iPhone

THE SCEPTIC

The upgrade treadmill has no summit — the hobby's own veterans say so, on the record

THIS SPECIAL

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FROM THE DESK

"For some years I have been afflicted with the belief that flight is possible to man."

— Wilbur Wright, letter to Octave Chanute, 13 May 1900¹

He wrote it as a confession, half-embarrassed, to a stranger whose opinion he respected. This issue is about the people who caught the same affliction a century later and never left the ground — MSFS and X-Plane and DCS World (p7), the study-level Airbus that fails exactly like the real one (p9), the throttle quadrants and the room-filling simpits (p11), the volunteers who run a real-feeling air traffic control network for nobody's paycheck (p13) — and what it costs, in money and hours, to stay afflicted.

THE NUMBERS — AT A GLANCE

METRIC	FIGURE
VATSIM active members (6 mo.)	~190,000
VATSIM registered, all-time	~800,000
MSFS 2024, Steam peak concurrent	24,784
Hoggit (DCS) Discord members	34,816
AVSIM.com, monthly visits	568,930
FlightSimExpo 2026 attendance	3,000+
Standard DCS World module	US\$80
PilotEdge, paid ATC network	US\$19.95/mo

Figures as researched for this edition, Aug 2026; see the desks named for full sourcing.²

WHY THIS ISSUE EXISTS

A reader flagged an article in chat and asked for the full treatment. Unlike the daily book, this special runs no news desks, no Diary, no Rabbit Hole — all 27 pages are eleven chapters on one subject.³

SOURCES ¹ Wilbur Wright to Octave Chanute, 13 May 1900; Octave Chanute Papers, Manuscript Division, Library of Congress. ² VATSIM Radar / VAT-STATS network statistics, Aug 2026; Steam concurrent-player trackers (Microsoft Flight Simulator 2024, app 2537590); Hoggit Discord member count via discord.com/invite/hoggit; avsim.com traffic via SEMrush, Dec 2025; FlightSimExpo 2026 (Saint Paul RiverCentre) attendance per AOPA and organiser reporting; DCS module and PilotEdge pricing from digitalcombatsimulator.com and pilotedge.net, Aug 2026. ³ Reader note logged via the ↗ special-edition request feature, 4 Aug 2026.

THE INVITATION · CHAPTER ONE

The Hobby That Starts With a Free Download

Home-cockpit flight simulation is three different communities pretending to be one hobby — a photoreal world, a physics engine, and a war game — and all three have spent the 2020s getting harder to tell apart from the real thing.

Nobody drifts into this hobby by accident. Somewhere between the free demo flight and the first real-money purchase, a specific thing happens: a photoreal render of an actual street loads outside the cockpit window, or a study-level Airbus refuses to start because a checklist item got skipped, or a friend's DCS module puts a functioning HUD in front of them for the first time — and the ordinary answer to "why would you pay for a flight you're not actually taking" stops applying.¹ What follows, for a meaningful minority, is a fairly serious hobby: real yokes and throttles bought with real money, payware aircraft priced like a decent dinner, hundreds of hours spent learning systems nobody will ever test them on, and — at the far end — rooms converted into single-purpose cockpits that never leave the house.

"Home-cockpit flight simulation" is the umbrella term for what is really three overlapping but distinct

scenes. There is the civil/general-aviation world built around Microsoft Flight Simulator and Laminar Research's X-Plane, which chases visual and systems realism in ordinary aircraft — a Cessna, an A320, a 747 — flown VFR for the view or IFR for the procedure.² There is the combat-sim world built around Eagle Dynamics' DCS World, which chases weapons-and-avionics fidelity in fast jets and helicopters, and treats "realistic" as a synonym for "hard." And there is the online air-traffic layer — chiefly the volunteer network VATSIM — that exists to make either of the first two feel like a shared world instead of a solo screensaver, by putting a real human being with a real headset on the other end of the radio.³ A given enthusiast might live in one of these worlds or in all three; the gear, the vocabulary and the running arguments differ more than outsiders expect.

24,784

MICROSOFT FLIGHT SIMULATOR 2024'S ALL-TIME PEAK CONCURRENT PLAYERS ON STEAM ALONE — A STEAM-ONLY FIGURE THAT UNDERCOUNTS THE REAL AUDIENCE, SINCE THE GAME ALSO SELLS ON XBOX AND THROUGH GAME PASS.⁴

Nobody signs up to file an IFR flight plan to a airport they will never visit. They sign up because a real street loaded outside the window, and for a second it looked like theirs.

THE CHATTER — HOW PEOPLE DESCRIBE THE HOOK

Across AVSIM's forums and r/flightsim, the origin story recurs in a handful of shapes: a relative's old copy of Flight Simulator on a childhood PC; a YouTube video of MSFS's Bing Maps-derived photogrammetry rendering a poster's actual neighbourhood; a DCS friend's F/A-18 startup sequence, all switches and no menus. What the accounts share is the moment the software stopped feeling like a game with an aviation skin and started feeling like a genuine, if incomplete, second cockpit.⁵

THREE COMMUNITIES

ONE UMBRELLA TERM

NO FINISH LINE

SOURCES 1 Community origin-story pattern as characterised across AVSIM and r/flightsim threads, Aug 2026 research. 2 Microsoft Flight Simulator (flightsimulator.com); Laminar Research X-Plane (x-plane.com) product positioning. 3 VATSIM.net, "Becoming a Pilot/Controller" documentation. 4 Steam concurrent-player trackers for Microsoft Flight Simulator 2024 (app 2537590), Aug 2026; cross-platform (Xbox/Game Pass) reach not captured by Steam-only trackers. 5 Community-sentiment synthesis, AVSIM.com and r/flightsim, Aug 2026 — paraphrased, not quoted verbatim.

THE INVITATION · SCALE

A Bigger World Than It Looks From Outside

The professional flight-simulator industry and the hobbyist home-cockpit scene are frequently quoted as if they were one market. They are not, and mixing them up flatters both.

Market-research reports on "the flight simulator market" — sized at roughly US\$7.2–7.6 billion for 2025–2026 by several industry trackers — are describing the commercial and military training industry: full-motion Level-D simulators, airline type-rating devices, defence contracts.¹ That figure has essentially nothing to do with a hobbyist buying a Honeycomb yoke. The consumer home-cockpit scene has no equivalent single market-size number that this issue could verify and print honestly — no analyst tracks HOTAS unit sales the way one tracks smartphones — so the only honest way to size it is by counting communities and platforms, not dollars.²

By that measure it is genuinely large. VATSIM, the volunteer online air-traffic network at the centre of

US\$7.2–7.6B

THE 2025–2026 "FLIGHT SIMULATOR MARKET," PER INDUSTRY TRACKERS — A FIGURE DESCRIBING FULL-MOTION AIRLINE AND DEFENCE TRAINING SIMULATORS, NOT THE HOBBY THIS ISSUE COVERS.

Chapter Six, counts roughly 190,000 members active in the past six months out of some 800,000 registered since 2001.³ AVSIM.com, the oldest surviving hobbyist hub, logged 568,930 visits in December 2025 alone.⁴ The DCS World combat-sim community's home base, the r/Hoggit-linked Discord, carries 34,816 members.⁵ And FlightSimExpo — the community's own trade show, not an industry one — drew more than 3,000 attendees and 70 exhibitors to Saint Paul, Minnesota in June 2026, part of almost 10,000 cumulative attendees since the show began in 2018.⁶ None of that is niche by hobby standards; it is a mechanical-keyboard-sized or a sim-racing-sized world, not a folk-craft-sized one.

THE CHATTER — BEING MISTAKEN FOR THE AIRLINE INDUSTRY

Enthusiasts occasionally note, with some irritation, when press coverage cites "the flight simulator market's billions" alongside a story about home-cockpit hobbyists — the two numbers describe entirely different industries, and the community itself is usually the first to point that out.⁷

THE SCEPTIC

The "just download the free demo" pitch undersells what serious participation costs. A recurring Steam-forum complaint about Microsoft's own add-on catalogue put it plainly: getting "everything" in Microsoft Flight Simulator would run into the hundreds of dollars before a single piece of dedicated hardware is bought — and one hardware-focused thread estimated a competent pedal-plus-HOTAS setup alone at roughly US\$1,000.⁸ The entry point is free. The hobby, taken seriously, is not — a tension this issue returns to directly in Chapter Eight ([p17](#)) and Chapter Eleven ([p23](#)).

WHERE THE REST OF THIS ISSUE GOES

Ch. 2–3 — the machines. A history of the four surviving simulators and where each one stands today.

Ch. 4–5 — the objects. The study-level aircraft that model failure as faithfully as flight, and the hardware bought to fly them.

Ch. 6–7 — the people. The volunteer air-traffic network, and the forums, Discords and channels where the culture actually lives.

Ch. 8–9 — the costs and the war games. What it really costs at every tier, and the combat-sim/VR wing of the hobby.

Ch. 10–11 — home and honesty. A Singapore angle taken honestly (thin, and said so), a glossary, the money pits the community itself names, and a real 90-day way in.

SOURCES 1 Flight simulator market-size estimates (professional/training/defence segment), Mordor Intelligence, Grand View Research, Fortune Business Insights and Precedence Research, 2025–2026 editions — commercial/military training market, distinct from the hobbyist consumer scene. 2 No verified consumer-hardware market-size figure exists for the home-cockpit hobby specifically; held rather than estimated. 3 VATSIM Radar / VAT-STATS network statistics, Aug 2026. 4 avsim.com traffic estimate via SEMrush, Dec 2025. 5 Hoggit Discord member count, discord.com/invite/hoggit, Aug 2026. 6 FlightSimExpo 2026 (Saint Paul RiverCentre, 12–14 June), attendance and exhibitor counts per AOPA reporting and organiser figures; cumulative attendance since 2018 per FlightSimExpo's own history. 7 Community commentary on training-vs-hobby market conflation, AVSIM/r/flightsim, Aug 2026 research — paraphrased. 8 Steam Community discussion threads, Microsoft Flight Simulator (2020) and Flight Simulator 2024 forums, community estimates on add-on and hardware spend, accessed Aug 2026 — forum sentiment, not a vendor figure.

CHAPTER TWO · ORIGINS

From an Apple II Program to a Server-Crashing Relaunch

Microsoft's *flight simulator* is the second-oldest software product still on sale under its original name — and it spent 2009 to 2020 effectively dead.

Bruce Artwick was writing magazine articles about 3D flight-simulation graphics in 1976 when a reader response convinced him there was a market for the real thing; he founded subLOGIC to build it.¹ SubLOGIC's Flight Simulator shipped for the Apple II in January 1979, and in 1982 the company licensed a CGA-graphics IBM PC version to Microsoft, which released it that November as Microsoft Flight Simulator 1.00 — the start of an unbroken product line that, forty-four years later, is still called Flight Simulator.² Artwick left subLOGIC and kept developing later Microsoft versions through his own studio; the series passed through FS3.0 (1988) and a string of successors up to Flight Simulator X (FSX) in 2006, built by Microsoft's in-house ACES Studio.³

Then it stopped. In January 2009, amid roughly 5,000 job cuts during the financial crisis, Microsoft closed ACES Studio outright — ending first-party development on the franchise with no announced successor.⁴ Microsoft's one attempt to revive the brand, a free-to-play reboot simply titled *Flight* in 2012, was pulled from sale within months after a hostile reception; one contemporary review called it "an insult to the simulation community," citing thin content, a cash-shop structure and — fatally for a hobby built on customisation — no support for third-party add-ons.⁵ For the next several years, the world's most recognisable flight simulator brand had no active product at all.

14 yrs

THE GAP BETWEEN FLIGHT SIMULATOR X (2006) — THE LAST RELEASE BEFORE MICROSOFT SHUT DOWN ITS IN-HOUSE STUDIO — AND THE GROUND-UP 2020 RELAUNCH THAT FINALLY REPLACED IT.⁶

Microsoft's own team had planned for 200,000 launch users. The real number overwhelmed the servers so badly the company had to explain itself publicly — a bad-problem-to-have that told you the hiatus had not killed the demand at all.

THE CHATTER — WHAT "IT'S BACK" FELT LIKE

The 18 August 2020 launch of the Asobo-built *Microsoft Flight Simulator* is the closest thing this hobby has to a shared origin memory for its current generation: satellite-derived photogrammetry rendering real streets and real weather in real time, on a scale nobody's home rig had shown before. Microsoft later admitted it had drastically underestimated demand, planning capacity for around 200,000 users and getting overwhelmed on day one — a launch-week failure mode the community read, correctly, as proof the hiatus had been starving a real appetite, not serving a shrinking one.⁷

1979 2009 2020

SOURCES 1 Bruce Artwick biography and subLOGIC founding, Wikipedia/aerosimulations.com history compilation, cross-checked Aug 2026. 2 subLOGIC Flight Simulator (Apple II, Jan 1979) and Microsoft Flight Simulator 1.00 (IBM PC, Nov 1982), per Microsoft/subLOGIC licensing history. 3 Bruce Artwick Organization and FS3.0 (1988) through Flight Simulator X (2006, ACES Studio) release lineage. 4 ACES Studio closure, 22–26 Jan 2009, amid Microsoft's ~5,000-role reduction; theprohack.com and contemporary reporting. 5 *Microsoft Flight* (2012), free-to-play reboot; withdrawn from sale within months; contemporary review calling it "an insult to the simulation community" (no third-party add-on support, cash-shop DLC). 6 Gap calculated FSX (2006) to Microsoft Flight Simulator (2020); OnMSFT.com franchise history. 7 Tom's Hardware, "Microsoft only planned for 200,000 users for Flight Simulator launch," 2020/2026 retrospective reporting on the day-one database overload.

CHAPTER TWO · THREE OTHER LINEAGES

While Microsoft Slept, Three Other Simulators Grew Up

X-Plane built its reputation on physics, not scenery. DCS World grew out of a Russian fighter-jet studio. And the "other" Microsoft simulator never actually died — it just stopped being for hobbyists.

Austin Meyer built the software that became X-Plane in the mid-1990s as a college project, distinguishing it from the start by how it calculates flight rather than how it renders the ground: X-Plane models an aircraft using blade-element theory, breaking wings, propellers and control surfaces into small segments and computing the aerodynamic force on each one, rather than looking up a pre-built performance table for that specific airframe.¹ The practical effect is that X-Plane can produce a plausible flight model for an aircraft nobody has flown before it exists — which is why the FAA has certified X-Plane-based devices as approved training aids (PCATD, BATD and AATD classes) in a way no purely visual simulator can claim.² Laminar Research, Meyer's company, has kept X-Plane's civil/GA focus distinct from Microsoft's scenery-first approach for three decades; X-Plane 12, the current version, is mid-way through a 2026 roadmap that includes a new avionics-panel framework (Q3) and a rebuilt flight model covering rotor dynamics, ground physics and wake turbulence (Q4).³

Eagle Dynamics traces a completely different line. Founded in 1991 by Nick Grey and Igor Tishin with a Moscow studio, it built its name on Su-27 Flanker (1995) — putting a real Soviet fighter's systems in front of Western PCs for the first time — then Flanker 2.0 (1999) and Lock On (2003), before starting the Digital Combat Simulator series in 2008 with the Ka-50 helicopter module Black Shark.⁴ DCS World, the free core platform that followed, is now sold as one free environment plus separately purchased aircraft modules — a business model this issue examines closely in Chapters Five and Eight. And Lockheed Martin's Prepar3D has a lineage no other simulator can claim: when Microsoft closed ACES Studio in 2009, Lockheed Martin bought the intellectual property for Microsoft ESP — the commercial-licensing spin-off of the FSX engine — and relaunched it as Prepar3D in 2010.⁵ It is, quite literally, Flight Simulator X's orphaned commercial sibling, still actively used for institutional training (a UK Air Cadets simulator donation ran as recently as February 2026) even though it never became a hobbyist mainstay the way its cousin did.⁶

THE CHATTER — THE ONE NOBODY MENTIONS

Ask a random flight-sim forum thread to name "the simulators" and Prepar3D rarely comes up unprompted — a striking absence for a product with a direct FSX bloodline. When it is mentioned, it is almost always by someone doing institutional or scenery-design work, not a hobbyist choosing where to spend a Saturday.⁷

2010

THE YEAR LOCKHEED MARTIN'S PREPAR3D LAUNCHED — BUILT FROM THE SAME MICROSOFT ESP CODEBASE ACES STUDIO HAD BEEN SELLING COMMERCIALLY BEFORE ITS CLOSURE THE YEAR BEFORE.

THE SCEPTIC

It is tempting to read this as four simulators competing on a level field. They are not. Prepar3D survives mainly because institutional training buyers value its licensing terms and stability guarantees over hobbyist features — an EULA historically restricted non-commercial home use in ways MSFS and X-Plane never were — so its continued existence says more about enterprise procurement than about consumer demand.⁸ Treat "four simulators" claims (including this issue's) as four different businesses that happen to share a genre, not four contestants in one race.

SOURCES 1 Austin Meyer/X-Plane origin and blade-element-theory flight model, x-plane.com press kit; General Aviation News, "Take off with X-Plane." 2 X-Plane FAA certification (PCATD/BATD/AATD device classes), x-plane.com. 3 Laminar Research 2026 roadmap (Panel Graphics API/avionics framework Q3 2026; flight-model rework Q4 2026), FlightSimExpo 2026 announcements via Threshold/ThresholdX.net. 4 Eagle Dynamics founding (1991, Nick Grey/Igor Tishin); Su-27 Flanker (1995), Flanker 2.0 (1999), Lock On (2003), DCS: Black Shark (2008) — company history compilations cross-checked against Wikipedia. 5 Prepar3D origin: Microsoft ESP licensed to Lockheed Martin following the 2009 ACES closure; Prepar3D announced May 2010, v1 Nov 2010 — key.aero, "Lockheed Martin Prepar3D: The story so far." 6 Lockheed Martin, RAF Air Cadets Prepar3D donation, Feb 2026 press release. 7 Community discussion pattern re: Prepar3D's low hobbyist mindshare, AVSIM Prepar3D forum, Aug 2026 research — characterised, not quoted. 8 Prepar3D historical EULA/commercial-licensing terms as widely discussed on AVSIM's Prepar3D forum, cross-checked against Lockheed Martin's own licensing pages.

CHAPTER THREE · THE PLATFORMS IN 2026

Three Simulators, Three Different Ideas of "Realistic"

Ask which flight simulator is "the best" and the honest answer is that the question assumes they are competing for the same prize. They are not.

Microsoft Flight Simulator 2024 launched in November 2024 as a paid successor rather than a free update to the 2020 title, and by late July 2026 it had sold an estimated 611,200 copies on Steam alone — a figure that, again, excludes Xbox and Game Pass players entirely and therefore understates Microsoft's actual reach by an unknown but probably large margin.¹ On a given day in August 2026 it holds roughly 5,000–6,000 concurrent Steam players, a stable base fluctuating by only a few per cent month to month — not a game in decline, but not the phenomenon its 2020 predecessor's launch-week chaos suggested either.² X-Plane 12 occupies a smaller but fiercely loyal niche:

contemporary reviews describe its default aircraft as a genuine step up in quality, "easily on par with many payware offerings" on rival platforms, and Laminar Research continues to lean into the FAA-recognised precision of its physics model rather than chase MSFS's satellite-photogrammetry scale.³ DCS World, meanwhile, has grown into a genuine company rather than a hobby project underneath its Cold-War-jet exterior: Eagle Dynamics reported roughly 190 employees in 2024 and has relocated its corporate headquarters from Moscow to Villars-sur-Glâne, Switzerland, even as its creative roots and much of its community remain Russian-speaking and Russia-adjacent.⁴

190

EAGLE DYNAMICS' APPROXIMATE EMPLOYEE COUNT IN 2024 — A COLD-WAR-JET HOBBY PROJECT THAT HAS GROWN INTO A GENUINE, HQ-RELOCATED COMPANY.

MSFS

BREADTH & SPECTACLE

The whole Earth, rendered — real weather, real air traffic density approximations, satellite-derived scenery, VR support out of the box. The pitch is scale: fly anywhere, in anything from a paraglider to an A380, and have it look plausible on day one.

X-PLANE

PHYSICS-FIRST PRECISION

Blade-element modelling means the flight characteristics of an aircraft are calculated, not looked up — which is why regulators, not just hobbyists, trust it. The pitch is correctness: if it flies wrong in X-Plane, that is treated as a bug worth fixing, not an acceptable approximation.

DCS WORLD

SYSTEMS & WEAPONS DEPTH

One theatre of war, a handful of aircraft, modelled switch-by-switch and weapon-by-weapon. The pitch is difficulty as a feature: a cold-start startup procedure that takes fifteen real minutes is the point, not friction to be streamlined away.

THE CHATTER — READING THE ROADMAPS AS TEA LEAVES

Enthusiasts track each platform's public roadmap the way stock analysts track earnings calls: Laminar Research's 2026 avionics-framework and flight-model announcements were parsed line by line on X-Plane forums within hours, largely to answer one recurring question — is this the update that finally closes the visual gap with MSFS without giving up the physics-model advantage that is X-Plane's whole reason to exist.⁵

SOURCES 1 Microsoft Flight Simulator 2024 Steam sales estimate (~611.2K copies), late July 2026, via SteamDB-derived trackers (playtracker.net, stmstat.com). 2 Steam concurrent-player data for Microsoft Flight Simulator 2024 (app 2537590), Aug 2026, via tracker.gg and steamcharts.com-derived sources; month-on-month variance ~3.7%. 3 X-Plane 12 default-aircraft quality assessment, key.aero and MKAU Gaming reviews, cross-checked. 4 Eagle Dynamics employee count (~190, 2024) and Villars-sur-Glâne, Switzerland headquarters, per Wikipedia company-infobox sourcing, cross-checked against Eagle Dynamics' own corporate filings references. 5 X-Plane community roadmap discussion pattern, xplanereviews.com and X-Plane forums, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER THREE • THE ARGUMENT THAT NEVER SETTLES

"Which One Should I Buy First?" Is the Hobby's Oldest Thread

The honest answer the community gives is almost always "it depends what kind of flying you actually want to do" — which is a more useful answer than it sounds.

Because the three main platforms optimise for different things, "which simulator should I buy" threads on AVSIM and r/flightsim rarely produce a single winner — they produce a triage. Someone who wants to fly a Cessna over their own house, in real weather, gets pointed at MSFS. Someone chasing procedural accuracy for real-world flight training or a home-cockpit build modelling one specific type gets pointed at X-Plane, partly on the strength of its regulatory pedigree. Someone who wants to fight, not commute, gets pointed at DCS World, with the caveat that its per-aircraft pricing (Chapter Eight) means "trying DCS" and "owning the module you actually want" are two different financial commitments.¹ Prepar3D rarely comes up in these threads at all outside professional-training contexts, for the licensing reasons noted in Chapter Two.

A genuine 2026 wrinkle worth naming honestly: Microsoft's console and Game Pass distribution gives

MSFS a scale advantage the Steam numbers above cannot show, which is part of why it dominates casual mindshare even among people who have never flown a study-level aircraft in it. That is a distribution win, not necessarily a simulation-quality win — and X-Plane and DCS loyalists say so, often and pointedly, in the same threads.²

Owning more than one platform, rather than picking a winner, is quietly the community's actual endpoint. It is common for a serious enthusiast to keep MSFS installed for weekend sightseeing, X-Plane for a specific study-level type they are learning properly, and DCS World's free core plus one module for evenings that call for combat rather than commuting — treating the "which simulator" question as a false choice rather than a real one, once the budget allows it.

3

THE NUMBER OF SEPARATE SIMULATOR ECOSYSTEMS A GENUINELY COMMITTED ENTHUSIAST TYPICALLY ENDS UP OWNING, PER THE COMMUNITY'S OWN "WHICH ONE" GUIDANCE THREADS — NOT THE ONE THIS ISSUE'S COVER TEASER IMPLIES.

The three simulators are not rivals so much as three different hobbies that happen to share the word "flight" — which is exactly why the "which one is best" argument never actually ends.

THE CHATTER — THE COMMUNITY'S OWN TRIAGE

Distilled from recurring AVSIM and r/flightsim guidance threads: buy MSFS if the goal is "fly anywhere, look at the world"; buy X-Plane if the goal is "learn one aircraft's systems precisely, maybe toward a real rating"; buy DCS World's free core (and exactly one module) if the goal is "fight, not commute" — and expect to eventually own more than one platform if the hobby actually sticks.³

THE SCEPTIC

MSFS's install-base numbers get quoted as a proxy for "best simulator" more often than its critics think is fair. A platform bundled into Xbox Game Pass and marketed with Microsoft's console reach will always out-count a PC-only niche title on raw player count, regardless of relative simulation fidelity — a distinction X-Plane and DCS communities raise whenever cross-platform comparisons get made, and one this issue has tried to keep visible rather than flatten into a single leaderboard.⁴

SOURCES 1 Community platform-selection guidance pattern, AVSIM.com and r/flightsim, Aug 2026 research, paraphrased. 2 Microsoft Flight Simulator Xbox/Game Pass distribution vs Steam-only tracking, cross-platform reach acknowledged but not separately quantified by this issue's sources. 3 Community triage pattern as above; not a verbatim quotation. 4 Community pushback on install-base-as-quality-proxy, as recurring in cross-platform comparison threads, AVSIM/r/flightsim, Aug 2026.

CHAPTER FOUR · THE PAYWARE AIRLINERS

Built to Fail Exactly Like the Real Thing

The most expensive add-ons in the hobby are prized not for what they do right, but for how faithfully they break when a real aircraft would.

"Study-level" is the hobby's term for an add-on aircraft detailed enough to be studied, not just flown — every switch wired to a simulated system, every failure mode modelled, every checklist item load-bearing. Three payware developers currently define the category on Microsoft's platform. PMDG, the longest-established of the three, built its reputation on the 737 line with what reviewers call exceptional systems coding and reliable performance.¹ Fenix Simulations released its A320-200 CFM on 19 May 2022 to a reception one contemporary review summarised as "one of the best products released for any flight simulator" — modelling essentially every switch, circuit breaker and failure interaction across the aircraft's electrical, hydraulic and pneumatic systems — while iniBuilds

has pushed its own A310 and A320neo lines into the same territory.² Fenix and iniBuilds' A310 are both now built on computational fluid dynamics (CFD) for their flight models, a technique previously reserved for real aerospace design work, rather than the older method of hand-tuning performance tables to match published figures.³

None of this is free, and none of it is casual. A study-level airliner from any of the three typically runs US\$70–90, before the scenery, liveries and utility add-ons enthusiasts stack around it — and the appeal is explicitly the difficulty, not despite it. A cold-and-dark A320 startup done "properly," following the same flow a real first officer would use, takes the better part of ten minutes before the aircraft is ready to taxi. The systems cult prizes exactly that friction.

2,710

FREE ADD-ONS BUILT FOR FLYBYWIRE'S OPEN-SOURCE A32NX ON FLIGHTSIM.TO ALONE — LIVERIES, CHECKLISTS, COCKPIT TWEAKS — THE SCALE OF COMMUNITY LABOUR THAT HAS GROWN AROUND A SINGLE FREEWARE AIRLINER.⁴

The aircraft is not admired for what it does easily. It is admired for how convincingly it can go wrong.

THE CHATTER — THE PMDG/FENIX RIVALRY

Comparative reviews of the two developers' flagship airliners recur constantly, and the community's own verdict has shifted over time: several contemporary write-ups now rank Fenix's A320 systems depth ahead of PMDG's 737 line, a genuine change from the years when PMDG's name alone settled the argument.⁵

THE SCEPTIC

Systems depth this granular is also fragile: a single incorrect failure-logic interaction can make an otherwise superb aircraft feel worse than a simpler one, and study-level developers' own changelogs are full of quiet fixes to exactly this kind of edge case, update after update.

STUDY LEVEL CFD FLIGHT MODELS \$70–90 EACH

SOURCES 1 PMDG 737 reputation and systems-coding quality, comparative add-on-vendor reviews via Microsoft Flight Simulator community forums, Aug 2026 research. 2 Fenix Simulations A320-200 CFM release (19 May 2022) and reception, simFlight and Threshold reviews; Twinfinite/Key.aero cross-checks. 3 CFD-based flight models: Fenix A320 and iniBuilds A310, per PMDG community forum discussion ("Both the Fenix and the iniBuilds A310 use CFD now") and iniBuilds' own product materials. 4 FlyByWire A32NX add-on count, flightsim.to affiliation page, Aug 2026; FlyByWire A32NX released under GNU GPLv3. 5 Comparative PMDG/Fenix reception discussion, Stormbirds.blog and Microsoft Flight Simulator community forums, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER FOUR · A CONTESTED TERM

Nobody Actually Owns the Definition of "Study Level"

And the claim that any of this substitutes for real training does not survive contact with the pilots who actually fly the real aircraft.

"Study level" has no certifying body and no fixed checklist — it is a community label, applied inconsistently, and AVSIM's own forums have run threads asking, plainly, "study 'level,' what is and what's not?" without landing on a settled answer.¹ Depth of systems modelling, accuracy of failure states, fidelity of the flight model and even cockpit texture resolution all get folded into the term by different people, which is why marketing copy leans on it more confidently than the community actually agrees with.

The bigger honesty question is whether any of this crosses over into real pilot training, and the evidence says: informally, yes; formally, no. Real Airbus pilots do use the Fenix A320 to stay procedurally sharp between simulator sessions or rehearse a flow before a check ride — that use case shows up openly in enthusiast and pilot-adjacent forums.² But at least one detailed review of the same aircraft concluded plainly that "real world Airbus pilots should look elsewhere if seeking training-oriented software," and

no consumer study-level add-on — on any of the three simulators covered in this issue — carries FAA or equivalent regulatory training credit; that status is reserved for certified devices (X-Plane's own certified professional builds among them, see Chapter Two) built and audited to a different standard entirely.³ The honest claim is "excellent procedural rehearsal for enthusiasts and off-duty familiarity for real pilots." The claim to avoid is "training device."

Chapter Nine (p19) will complicate this picture further: DCS World, a purely hobbyist combat simulator with none of study-level civil aviation's training pretensions, has a more credibly documented military training use case than any product in this chapter. The lesson is not that combat sims are more serious than civil ones — it is that training value turns out to be less correlated with marketing claims than either side of the hobby assumes.

THE CHATTER — DEFINING IT ANYWAY

Pressed to define "study level" despite the lack of a formal standard, the community's rough consensus lands on three tests: does every switch do something a real system would recognise; does the aircraft fail the way the real one fails, not just stop working; and does the manual read like an aircraft's operating handbook rather than a game's tutorial. No single add-on this issue researched passes all three tests perfectly — the term describes a direction, not a finish line.⁴

THE SCEPTIC

Marketing copy for study-level add-ons routinely implies a closeness to "the real thing" that the products themselves do not claim in their fine print, and that at least one independent review explicitly contradicted for the Fenix A320's training value. The systems depth is real and often extraordinary; the training-equivalence framing that sometimes surrounds it in community discourse is not something this issue can verify, and prints it here only to flag it, not to endorse it.⁵

\$70–90

THE TYPICAL PRICE OF ONE STUDY-LEVEL AIRLINER FROM PMDG, FENIX OR INIBUILDS — BEFORE THE SCENERY, LIVERIES AND UTILITY ADD-ONS ENTHUSIASTS STACK AROUND IT.

STUDY LEVEL

CONTESTED TERM

NOT A TRAINING CREDIT

SOURCES 1 AVSIM Community, "Study 'level,' what is and what's not?" forum thread, Microsoft Flight Simulator (2020/2024) section. 2 Informal real-pilot use of study-level add-ons for procedural familiarity, as discussed in pilot-adjacent forum threads (Pilots of America, AVSIM), Aug 2026 research — anecdotal, not a formal training claim. 3 Fenix A320 review assessment on training suitability ("real world Airbus pilots should look elsewhere"), simFlight review, cross-checked; FAA training-device certification (PCATD/BATD/AATD) reserved for certified devices per Chapter Two sourcing. 4 Community "study level" definition consensus, AVSIM/r/flightsim, Aug 2026 research — characterised, not a formal standard. 5 Editorial assessment; no vendor claims formal training-equivalence for consumer study-level add-ons as of Aug 2026.

CHAPTER FIVE • THE GEAR

From a US\$50 Stick to a Room With No Other Purpose

The hardware ladder has four rungs, and almost nobody who stays in this hobby for more than a year stops on the first one.

The entry rung is genuinely cheap. A Thrustmaster T.Flight Stick X lists at US\$49.99 as the standard "am I actually going to like this" purchase, and a step up to a proper combined HOTAS (hands-on-throttle-and-stick) unit like the T-Flight Hotas X still lands under US\$75; Thrustmaster's cross-platform HOTAS One runs US\$99.99.¹ This is where the large majority of people who ever open a flight simulator remain, and there is no shame the community attaches to staying there — a stick-and-throttle combo is genuinely enough to fly and enjoy every simulator in this issue.

The mid tier is where "hobbyist" starts to mean something. Honeycomb Aeronautical's Alpha yoke

and Bravo throttle quadrant are priced at US\$299.99 each — call it roughly US\$550–600 as a bundle — and model a real GA-through-airliner throttle quadrant with autopilot controls, an annunciator panel and a trim wheel built in.² WinWing's Orion 2 HOTAS, aimed at the combat-sim side of the hobby, runs US\$400–500 for a faithful F/A-18-style stick-and-throttle set with metal internals and a genuine afterburner detent.³ Add a set of rudder pedals — MFG's Crosswind V3, the community's default recommendation, runs roughly US\$336–372 depending on the model — and a mid-tier setup comfortably clears US\$1,000 before any aircraft additions are purchased at all.⁴

The cost ladder (typical, USD)

Entry ~\$50–100



Mid ~\$1,000–1,300



High \$2,000–4,000+



Full simpit — open-ended



Hardware plateaus by the VIRPIL tier. Simpit builds have no agreed ceiling — multi-year projects are common (Ch. 8, p17).

Source: Thrustmaster/Honeycomb/WinWing/VIRPIL/MFG pricing, Aug 2026. Schematic.

Bar lengths are illustrative, not linear — the point is the gap between tiers, not exact proportion.

SOURCES 1 Thrustmaster T.Flight Stick X (\$49.99), T-Flight Hotas X (under \$75) and HOTAS One (\$99.99) pricing, manufacturer and retailer listings, Aug 2026. 2 Honeycomb Aeronautical Alpha Flight Controls and Bravo Throttle Quadrant, \$299.99 each, flyhoneycomb.com, Aug 2026. 3 WinWing Orion 2 HOTAS, \$400–500 full set, per boostedmedia.net and manufacturer listings. 4 MFG Crosswind V3 rudder pedals, ~\$336–372 (model-dependent), mfg-sim.com, Aug 2026.

CHAPTER FIVE · THE TOP RUNGS

Where the Ladder Stops Having a Top

VR headsets and full simpit builds are where the hobby stops resembling a purchase and starts resembling a hobby project with no fixed end date.

VIRPIL's controls sit at the top of the packaged-hardware tier: its CM3 Throttle is widely regarded among combat-sim buyers as one of the best standalone throttles available, but the products ship from Lithuania and Belarus, and a full VIRPIL stick-and-throttle setup easily clears US\$600 — before shipping and import costs that vary sharply by country.¹ Layer in a VR headset and the "high" tier gets genuinely expensive: Meta's Quest 3 gives most buyers the best balance of price and usability at US\$599 for the 512GB model, while headsets built for maximum instrument-panel clarity — the Pimax Crystal Super and Crystal Light, Varjo's XR-4 — trade wireless convenience for resolution good enough to actually read a glass cockpit's smaller dials, at a correspondingly premium price.²

And then there is the tier with no published price at all, because nobody buys it off a shelf. Full simpit builders construct physical cockpit shells — a real 737 or A320 layout, complete with printed or salvaged panel overlays — in spare rooms and garages; one widely covered UK build, by hobbyist David Naylor in Wombwell near Barnsley, replicated a 737 cockpit in his front room over sixteen months, sourcing components from manuals, YouTube tutorials and specialist suppliers.³ Companies including Opencockpits, CPflight and SimObsession sell the panel hardware that makes these builds possible without machining everything from scratch — but the projects remain fundamentally open-ended: there is always another overhead panel, another annunciator light, another system to wire in.

16 mos.

HOW LONG UK HOBBYIST DAVID NAYLOR SPENT BUILDING A FULL 737 COCKPIT REPLICA IN HIS OWN FRONT ROOM — SOURCED FROM MANUALS, YOUTUBE AND SPECIALIST SUPPLIERS, NOT A KIT.

THE BUYER'S LADDER — FOUR RUNGS

Entry, ~\$50–100. A single stick or basic HOTAS (Thrustmaster T.Flight range). Enough to fly and enjoy everything in this issue.

Mid, ~\$1,000–1,300. A dedicated yoke or HOTAS (Honeycomb, WinWing) plus rudder pedals (MFG Crosswind). Where "hobbyist" starts.

High, \$2,000–4,000+. Premium sticks/throttles (VIRPIL) plus a serious VR headset (Quest 3 through Pimax/Varjo). Diminishing returns begin here (see Chapter Eleven, p23).

Full simpit, open-ended. A dedicated room, real panel hardware, multi-year builds. No community-agreed ceiling — see Chapter Eight (p17) on cost.

THE CHATTER — "BUY ONCE, CRY ONCE"

The hardware forums' recurring advice to newcomers is to skip the cheapest mid-tier options entirely: buy the more expensive yoke or HOTAS once, the reasoning goes, rather than replace a budget unit within a year and pay twice. It is standard enthusiast-hardware advice repackaged for this specific hobby, and it is also, not coincidentally, advice that keeps the upgrade spending flowing.⁴

THE SCEPTIC

"Buy once, cry once" is sound advice and also a perfect justification for never stopping. Chapter Eleven's Sceptic section (p23) goes further into the community's own admissions about the upgrade treadmill this advice quietly feeds.

SOURCES 1 VIRPIL CM3 Throttle reputation and pricing (~\$600+ full setup), virpil-controls.eu and helisimmer.com review, Aug 2026; shipping origin Lithuania/Belarus per virpil-controls.us.com. 2 Meta Quest 3 (\$599, 512GB) and premium alternatives (Pimax Crystal Super/Light, Varjo XR-4), 2026 buyer's-guide roundups (boundbyflame.com, gamingpcguru.com), cross-checked. 3 David Naylor 737 home-cockpit build, Wombwell, Barnsley — BBC News (South Yorkshire), 16-month build timeline. Opencockpits, CPflight, SimObsession as representative panel-hardware suppliers, company sites. 4 Hardware-forum "buy once, cry once" advice pattern, AVSIM/r/flightsim hardware threads, Aug 2026 — paraphrased community norm, not a single quotation.

CHAPTER SIX · THE VOLUNTEER NETWORK

190,000 People Fly for Nobody's Paycheque

VATSIM is *what turns a solo flight-simulator session into something closer to a real airspace — one staffed entirely by people who signed up to do it for free.*

A flight simulator on its own is a very good solo experience. What it lacks — until an enthusiast connects to a network — is other traffic that behaves like real traffic, and air traffic control that sounds and functions like the real service. VATSIM, the Virtual Air Traffic Simulation network, solves this by routing simmers' aircraft and simmers' voices through a shared network: someone logs in as a controller at a real airport's tower or approach frequency, using real phraseology, and pilots fly to and from that airspace exactly as they would in the actual aviation system.¹ The network claims roughly 800,000 registered accounts across its history, with about 190,000 members active in the past six months — a serious, if long-tailed, population for a hobby built entirely on volunteer labour.²

VATSIM's founding is itself a small piece of flight-sim history. In the summer of 1997, an earlier online-ATC network called SATCO collapsed; two rival successor projects grew out of the resulting scramble, built around Jason Grooms' ProController and SquawkBox client software. One became VATSIM, formally organised in July 2001; the other tradition fed into IVAO, the International Virtual Aviation Organisation, founded as a Belgian nonprofit on 16 December 1998.³ Both remain active and free today, run entirely by volunteer staff, and both have spent a quarter-century building the procedural culture — real phraseology, real separation standards, real-feeling delay vectoring — that distinguishes this corner of the hobby from simply flying alone.

50 hrs

THE TYPICAL MINIMUM CONTROLLING TIME REQUIRED AT EACH VATSIM RATING BEFORE A TRAINEE CONTROLLER CAN BEGIN WORKING TOWARD THE NEXT ONE UP THE LADDER.⁴

Nobody pays these controllers, and nobody has to fly into their airspace. Both facts make the ones who show up every week a genuinely strange kind of dedicated.

The VATSIM ratings ladder

S1 — Student / Tower Trainee

S2 — Tower (after ~50 hrs)

S3 — Approach (after ~50 hrs)

C1 — Centre (after ~50 hrs)

Source: VATSIM UK/VATNZ, Aug 2026.

Senior (C3) and instructor (I1/I3) ratings sit above C1.

THE CHATTER — THE REGULARS' OWN SLANG

Long-serving VATSIM controllers develop a private vocabulary around their home ARTCC or FIR much like real controllers do — nicknames for recurring virtual pilots, running jokes about a particular fix or frequency, a house style for handling the network's inevitable share of confused newcomers. It reads, on the discussion boards, exactly like any volunteer organisation's inside culture.⁵

SOURCES 1 VATSIM.net, network structure and function ("Becoming a Pilot," network overview). 2 VATSIM Radar / VAT-STATS network statistics, Aug 2026 (~800,000 registered, ~190,000 active past 6 months). 3 SATCO collapse (1997) and the VATSIM/IVAO split, via fsnews.eu "Simcident Report" retrospective; VATSIM formed July 2001 per community history threads (VATSIM UK forum); IVAO founded 16 Dec 1998 per ivao.aero and Wikipedia. 4 VATSIM ATC rating progression (S1 → S2 → S3 → C1), approximately 50 controlling hours per step, per VATSIM UK "New Controller" and vatnz.net training documentation. 5 VATSIM controller-culture pattern, VATSIM UK/VATSIM community forum discussion, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER SIX · FREE VS. PAID

The One Network That Charges — and Why Some Pilots Pay It

PilotEdge sells the thing VATSIM cannot guarantee: that the controller will actually be there, every time, doing the job the same way.

PilotEdge is the hobby's one significant paid alternative: a US\$19.95-a-month subscription service (or under \$15/month billed annually) staffed by paid controllers rather than volunteers, explicitly aimed at real-world pilots seeking currency practice as much as at simulator enthusiasts.¹ Its own comparison materials make the trade-off explicit and unflattering to the free networks: volunteer platforms, PilotEdge argues, "cannot deliver reliable, repeatable experiences" because controller presence, controller quality, traffic density and even fellow pilots' competence are all unguaranteed on a hobbyist roster.² That is a vendor making its own case, not a neutral verdict — but it names a real structural difference rather than inventing one.

The rating ladder is what actually organises VATSIM's volunteer controller corps into something coherent despite that variability. A new controller starts as S1 (student/tower trainee); after roughly 50 hours of controlling time they can progress to S2 (full tower

authority); another 50 hours or so of controlling opens the path to S3 (TMA/approach, covering procedural work up to and including all tower positions); and a further roughly 50 hours leads to C1, the enroute or "centre" rating, with senior (C3) and instructor (I1/I3) ratings available to those who stay involved.³ Every step requires logged hours and, from S2 onward, a written and practical exam — a genuine apprenticeship structure for something nobody is paid to do.

IVAO runs a broadly parallel, separately administered rating structure for its own volunteer controllers and pilots, and the two networks' communities have historically overlapped less than an outsider might expect — most enthusiasts pick one network and build their reputation and rating history inside it, rather than splitting hours between both, since a controller's standing (and their region's roster of familiar regular pilots) does not transfer between the two.⁶

THE CHATTER — WHY PEOPLE ACTUALLY CONTROLLER

Ask a VATSIM controller why they spend an unpaid evening running a tower frequency and the honest answers cluster around two things: the genuine satisfaction of running a busy sequence well, and a community identity built around a specific ARTCC or region that functions much like a sports-club membership. It is volunteer infrastructure work wearing a hobby's clothes.⁴

THE SCEPTIC

PilotEdge's own marketing is the sharpest critique of VATSIM in print, and it is not disinterested: a paid competitor arguing that volunteer ATC is structurally unreliable is also making the case for its own subscription. Both things can be true — VATSIM's ATC coverage genuinely does thin out on off-peak hours and outside major fly-in events, and PilotEdge genuinely is a paid product built to profit from that gap.⁵

US\$19.95

PILOTEGE'S MONTHLY SUBSCRIPTION — THE ONE RECURRING HARDWARE- AND CONTENT-FREE COST THIS CHAPTER FOUND ANYWHERE IN THE ONLINE-ATC LAYER OF THE HOBBY.

VOLUNTEER

RATING LADDER

\$19.95/MO ALTERNATIVE

SOURCES 1 PilotEdge subscription pricing (\$19.95/mo, under \$15/mo annual), pilotedge.net, Aug 2026. 2 PilotEdge, "Comparison to Other Networks," pilotedge.net — vendor's own framing. 3 VATSIM ATC rating progression and hour requirements, VATSIM UK "New Controller" and vatnz.net training documentation, Aug 2026. 4 Controller motivation pattern as characterised across VATSIM community forum discussion, Aug 2026 research — paraphrased. 5 PilotEdge vs. VATSIM/IVAO comparative discussion, Pilots of America and AVSIM community threads, cross-checked against PilotEdge's own comparison page. 6 IVAO's parallel rating structure and community overlap pattern with VATSIM, per ivao.aero and AVSIM/community forum discussion, Aug 2026 research.

CHAPTER SEVEN • WHERE THE CULTURE LIVES

A Twenty-Six-Year-Old Forum Still Outdraws the Reddit

The hobby's centre of gravity is older, and more fragmented by platform tribe, than its social-media presence suggests.

AVSIM.com predates most of the discourse that now happens on Reddit and Discord — a nonprofit hub built around forums, a file library and product reviews, run by volunteers and funded by donations and advertising, covering Microsoft's simulator, Prepar3D and X-Plane under one roof.¹ Its scale is still real: 568,930 visits in December 2025 alone, per third-party traffic estimates, at a point in the hobby's life when most enthusiast communities have long since migrated entirely to Discord.² r/flightsim carries the broader civil/GA-sim conversation on Reddit, without a verifiable public subscriber figure this issue could confirm and print honestly; general-purpose flight-

sim Discord servers separately run into the thousands of members each.³

The combat-sim side has its own, more self-contained ecosystem, centred on Hoggit — a Reddit-born community that describes itself as "a laid back place for flight simmers of all skill levels to get together and fly online while learning DCS World modules," in deliberate contrast to more strictly regimented multiplayer DCS squadrons elsewhere.⁴ Hoggit's linked Discord carries 34,816 members and runs a standing "Tuesday Night Noob" session specifically to ease newcomers into multiplayer combat flying without getting shot down by veterans in the first five minutes.⁵

568,930

AVSIM.COM'S ESTIMATED VISITS IN DECEMBER 2025 — A 26-YEAR-OLD VOLUNTEER-RUN FORUM STILL PULLING NUMBERS MOST MODERN NICHE COMMUNITIES WOULD ENVY.⁶

The forum is older than most of its current readers' interest in aviation. It is also, measurably, still where a lot of them end up.

THE CHATTER — WHY THE OLD FORUM SURVIVES

AVSIM's own users tend to credit its file library and product-review archive over its discussion threads: two decades of add-on compatibility notes, install fixes and long-form reviews that Discord's ephemeral chat format was never built to preserve. A forum's searchability, not its liveliness, is what keeps pulling people back.⁷

THE SCEPTIC

AVSIM's traffic numbers are also propped up by search engines routing troubleshooting queries to a decade-old thread rather than a live, active community — a forum can be heavily visited without being heavily inhabited, and this issue could not separate the two from the traffic figure alone.

AVSIM HOGGIT R/FLIGHTSIM

SOURCES 1 AVSIM.com nonprofit structure and scope (MSFS, Prepar3D, X-Plane), avsim.com/forums about page. 2 avsim.com traffic estimate, SEMrush-derived, Dec 2025. 3 r/flightsim subreddit existence and general-purpose flight-sim Discord servers (several thousand members each, e.g. discord.com/invite/Fq3BfmH), Aug 2026 — no verifiable current subscriber count for r/flightsim found and printed; held rather than estimated. 4 Hoggit community self-description, wiki.hoggitworld.com. 5 Hoggit Discord member count (34,816) and "Tuesday Night Noob" event, discord.com/invite/hoggit, Aug 2026. 6 See note 2. 7 AVSIM user-value pattern (file library/review archive over live discussion), AVSIM.com forum structure and community discussion, Aug 2026 research.

CHAPTER SEVEN · THE GATHERING, THE SCREENS

One Trade Show, a Handful of Channels, and the Running Arguments

FlightSimExpo is the one place the whole hobby — civil, combat, hardware, ATC — occupies the same room at once.

FlightSimExpo is the community's own trade show rather than an industry one, and it is the rare event where MSFS pilots, X-Plane loyalists, DCS fliers and hardware builders all show up under one roof. The 2026 edition, held 12–14 June at the Saint Paul RiverCentre in Minnesota, drew more than 3,000 attendees and 70 exhibitors, with hands-on access to more than 60 simulator setups, part of almost 10,000 cumulative attendees across shows since the event began in 2018.¹ Manufacturers use it as a launch venue — Honeycomb Aeronautical has attended as a sponsor debuting new hardware, and SKALARKI Electronics unveiled a dual-seat A330 home-cockpit setup at a recent edition — which makes the show a genuine barometer of where the hardware side of the hobby is heading.²

On video, the picture is less tidy. VASAviation — built almost entirely around real air-traffic-control audio recordings rather than simulator footage — has built a 670,000-subscriber channel that draws heavily from the same aviation-obsessive audience as the sim community, even though it is not, strictly, a flight-sim channel.³ C.W. Lemoine, a real fighter pilot who posts as "Mover," occupies a similar adjacent space, reviewing DCS modules and military-aviation content with the authority of someone who has actually flown the aircraft being simulated.⁴ This issue did not find a verifiable, dominant "flight-sim YouTuber" in the way gaming genres often produce one breakout creator — the audience is real but fragmented across platform-specific channels, addon-review specialists and crossover aviation content, and no single figure commands the whole room.

THE CHATTER — THE RUNNING ARGUMENTS

Three arguments recur across every forum covered on this page, unresolved by design: whether a given payware add-on is worth its price relative to a free community alternative; whether a developer's promised update timeline should be trusted after a history of slipping; and — the oldest one — whether the platform someone didn't choose is secretly the better simulator. None of these get settled. That is, by most accounts, part of the appeal.⁵

THE SCEPTIC

A hobby this dependent on forum reputation and developer trust is also unusually exposed to hype cycles: a well-marketed add-on can draw pre-order money on the strength of screenshots and a developer's past reputation, months before independent reviewers can confirm whether the promised systems depth actually shipped. Chapter Eleven's Sceptic section (p23) names specific instances the community itself flags as cautionary.

60+

SIMULATOR RIGS, FROM BASIC HOTAS DESKS TO FULL MOTION PLATFORMS, ON THE SHOW FLOOR FOR HANDS-ON TRIAL AT FLIGHTSIMEXPO 2026 — THE ONE PLACE THIS HOBBY'S ENTIRE HARDWARE LADDER IS TOUCHABLE IN ONE AFTERNOON.

FLIGHTSIMEXPO

VASAVIATION

"MOVER"

SOURCES 1 FlightSimExpo 2026 (Saint Paul RiverCentre, 12–14 June), attendance/exhibitor figures per AOPA reporting and organiser materials; cumulative attendance since 2018 per FlightSimExpo history. 2 Honeycomb Aeronautical FSExpo sponsorship and product launches, flyhoneycomb.com news; SKALARKI Electronics A330 dual-seat cockpit, FlightSimExpo 2025 coverage. 3 VASAviation, 670,000 YouTube subscribers, channel focus on ATC audio, per channel statistics cross-checked Aug 2026. 4 C.W. Lemoine ("Mover"), fighter pilot and content creator, DCS/military-aviation content, per creator bio and channel content, Aug 2026. 5 Community discourse pattern, AVSIM/r/flightsim/Hoggit, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER EIGHT · WHERE THE MONEY GOES

The Hardware Is a One-Time Cost. The Content Is Not.

This is the one flight-simulator chapter with genuinely good news in it: two of the hobby's most-used networks are, and have always been, free.

Chapter Five priced the hardware ladder; this chapter is about what keeps recurring after the hardware is bought. On the civil/GA side, the recurring spend is content: a single study-level airliner from PMDG, Fenix or iniBuilds runs roughly US\$70–90, and enthusiasts who fly more than one aircraft type routinely own several — three or four study-level airliners is not an unusual collection, which puts a genuinely engaged MSFS or X-Plane user at US\$250–350 a year in aircraft alone, before scenery add-ons for specific airports are counted.¹ The community's own freeware ecosystem — FlyByWire's A32NX chief among it — exists in part as a direct answer to that cost, and it is genuinely excellent; it does not eliminate the pull toward payware, it just gives newcomers a real, no-cost option while they decide whether the hobby is for them.²

DCS World's economics run differently, and more contentiously. The core simulator and its default map are free; every additional aircraft module is a separate purchase, standard-priced at US\$80, with the F/A-18C Hornet specifically listed at \$79.99.³ Sales bring modules to roughly \$40, and Eagle Dynamics runs them often enough that community veterans wait rather than buy at full price — but the underlying complaint recurs on Steam's own discussion boards regardless: one 2024-era thread put it as "is it me or is DCS getting expensive," and another poster's recollection that module prices have risen three-to-four-fold over five years for some regions is a forum claim this issue could not independently verify against an official price-history dataset, and prints here flagged as such — a community perception, not a confirmed fact.⁴

US\$80

THE STANDARD PRICE OF ONE DCS WORLD AIRCRAFT MODULE — BEFORE THE SALE THAT, IN THIS COMMUNITY'S EXPERIENCE, USUALLY ARRIVES EVENTUALLY.⁵

Two of the biggest things in this hobby have never charged a cent. Almost everything else, eventually, does.

THE CHATTER — WAITING FOR THE SALE

Veteran DCS buyers openly advise newcomers never to buy a module at launch price unless they cannot wait: Eagle Dynamics runs sales often enough, community-shared spreadsheets tracking historical module discounts have become their own small cottage genre, used to time purchases against past sale calendars.⁶

THE SCEPTIC

"Wait for the sale" is good individual advice and a bad system-level signal: if buyers systematically delay full-price purchases, a developer's sustainable long-run price is arguably closer to the sale price than the list price — which raises the uncomfortable question of whether the \$80 sticker was ever really the honest number.

SOURCES 1 PMDG/Fenix/iniBuilds study-level airliner pricing (~\$70–90 each), per Chapter Four sourcing. 2 FlyByWire A32NX, free/open-source (GPLv3), flightsim.to and FlyByWire project materials. 3 DCS World free core + paid modules; F/A-18C Hornet at \$79.99, digitalcombatsimulator.com, Aug 2026. 4 Steam Community discussion threads, "Is it me or is DCS getting Expensive 2024," DCS: F/A-18C ED Forums ("80\$ for the F-18? really?"), and community price-increase claims, accessed Aug 2026 — forum sentiment, flagged as unverified against an official dataset. 5 See note 3; sale pricing (~\$40, 50% off) per recurring DCS sale-event reporting. 6 DCS sale-timing advice pattern and community discount-tracking spreadsheets, Steam Community and DCS forum discussion, Aug 2026 research.

CHAPTER EIGHT · THREE ILLUSTRATIVE YEARS

What a Year Might Actually Cost — Worked, Not Surveyed

No credible survey measures average annual spend across this hobby, so this issue built three honest, itemised sketches from the verified prices already in this book instead of inventing a number.

These are arithmetic, not data: nobody has published a representative survey of annual flight-sim spend, so rather than invent an average, this issue built three illustrative profiles directly from the per-item prices already sourced across Chapters Four, Five and this chapter. Treat them as worked examples of plausible spending shapes, not measured averages.

The pattern that holds across all three, worked or not, is the one the community itself names most often: the hardware plateaus — a HOTAS, a yoke, a set of pedals eventually stop needing replacement — but the content spend does not have a natural ceiling, because there is always another aircraft, another module, another airport add-on that a specific new feature or sale makes newly tempting.¹

THREE ILLUSTRATIVE YEARS (BUILT FROM THIS ISSUE'S OWN PRICES)

The MSFS completionist. Three to four study-level airliners a year at \$70–90 each (\$210–360), plus scenery add-ons — call it roughly \$300–500 in content, on top of hardware already owned.

The DCS collector. Two to three modules a year, mixing full price and sale pricing (\$40–80 each) — roughly \$150–240, plus the occasional campaign add-on pack.

The simpit builder. No natural ceiling. Panel hardware, printed overlays and replacement parts arrive in an ongoing trickle for as long as the build continues — which, per Chapter Five's example, can run well over a year.

THE SCEPTIC

All three of the sketches above are honest arithmetic and none of them is a measured average — a distinction this issue is making deliberately, because "flight sim costs \$X a year" headlines circulate in the hobby's own press without that caveat attached, and it is exactly the kind of unverifiable-but-plausible-sounding number the honesty standard this book holds itself to exists to prevent.²

Nobody surveyed this hobby's spending because nobody had to — every enthusiast already knows roughly what their own year cost, down to the dollar, because every purchase in it was a deliberate, individually remembered decision.

THE CHATTER — TRACKING IT ANYWAY

Some enthusiasts keep informal spreadsheets of their own flight-sim purchases, half as a budgeting discipline and half as a joke about the hobby's reputation — a self-aware admission that the running total is exactly the number a spouse or a bank statement might eventually ask about.³

3

ILLUSTRATIVE SPENDING PROFILES THIS CHAPTER BUILT — COMPLETIONIST, COLLECTOR, SIMPIT BUILDER — NONE OF THEM A MEASURED AVERAGE, ALL OF THEM ARITHMETIC ANYONE CAN CHECK AGAINST CHAPTERS FOUR AND FIVE.

WORKED, NOT SURVEYED

NO NATURAL CEILING

SOURCES 1 Recurring-spend pattern (hardware plateaus, content does not), synthesised from Chapters Four, Five, Eight and community discussion, Aug 2026. 2 Editorial note on the absence of a credible annual-spend survey for this hobby, Aug 2026 research; no such dataset located. 3 Informal personal-spend-tracking pattern, AVSIM/r/flightsim community discussion, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER NINE · THE WAR GAME THAT TRAINS REAL PILOTS

A-10 Pilots Actually Train in This, in VR

DCS World is a commercial hobby product with a genuine, documented military training use case — a claim this issue could verify, unlike some of the training claims in Chapter Four.

Chapters Two and Three covered Eagle Dynamics' history and DCS World's current standing; this chapter is about what the combat-sim end of the hobby actually does with it, and how far that has genuinely crossed into real aviation. The most striking documented case is real: A-10 Warthog pilots have used DCS World in VR headsets as an actual training aid, reported by the defence-focused outlet The War Zone — a case where a hobbyist product, not a certified training device, earned real operational use because of what its visual and systems fidelity, combined with VR's depth perception, could offer that cheaper alternatives could not.¹ That is a materially stronger training claim than anything in Chapter Four's civil-airliner section, and this issue is printing it because

the sourcing holds up, not because it fits a tidier narrative.

VR is the specific ingredient that makes this plausible. Pilots and reviewers converge on the same two benefits: genuine depth perception for close-formation and dogfighting work that a flat monitor cannot replicate, and a harder-to-quantify sense of "presence" — the scale of sitting inside an actual cockpit rather than looking at one — that one pilot described feeling in his stomach the first time he lifted a simulated Huey off the ground.² Chapter Five already priced the headsets that deliver this (p11); this chapter is about why the combat-sim community specifically treats VR as closer to mandatory than the civil-sim side does.

1

THE NUMBER OF REAL US AIR FORCE AIRFRAMES — THE A-10 THUNDERBOLT II — THIS ISSUE FOUND A CREDIBLY REPORTED CASE OF PILOTS TRAINING ON USING A COMMERCIAL, OFF-THE-SHELF COMBAT FLIGHT SIMULATOR.³

The switches are simulated. The report that a real pilot's stomach dropped the first time he lifted off in VR was not.

THE CHATTER — WHAT THIS MEANS TO THE COMMUNITY

The A-10 training story circulates constantly in DCS discussion as validation — proof, to a hobby often dismissed as "just a game," that the fidelity it obsesses over has real-world value. It is cited far more often than it is examined critically, which is exactly why this chapter tried to source it properly rather than repeat it on faith.⁴

THE SCEPTIC

One well-documented training use case is not evidence that DCS World generally substitutes for military simulators — the A-10 case involved a specific unit's specific supplementary use, not a service-wide training mandate, and the community's tendency to generalise from it deserves the same scepticism this issue applies to Chapter Four's civil-training claims. What can be said cleanly is narrower and still remarkable: a consumer combat simulator, bought off the shelf, got close enough to the real thing that a real squadron found it worth the licence fee.

A-10 THUNDERBOLT II

VR PRESENCE

SUPPLEMENTARY, NOT MANDATED

SOURCES 1 "A-10 Warthog Pilots Are Using The Digital Combat Simulator Video Game To Train In VR," The War Zone (twz.com), cross-checked Aug 2026. 2 VR presence/depth-perception benefits for dogfighting and formation flying, key.aero "Virtual Reality — Part 1"; pilot testimonial (Huey lift-off sensation) via Pilots of America community discussion. 3 See note 1. 4 DCS community discussion pattern citing the A-10 VR training case, Hoggit/DCS forums, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER NINE · THE CAMPAIGN WRITERS

The People Who Write DCS's Missions Talk to Real Pilots Too

The single-player campaign market inside DCS World is small, name-driven, and unusually well-sourced for a hobby product.

DCS World's free core comes with one map and no story; the scripted single-player campaigns that give it narrative structure are almost entirely third-party work, and the field is dominated by a small number of recognised names. Baltic Dragon, the most prolific, has built a catalogue including *Raven One*, *Raven One: Dominant Fury* and *Iron Flag*, and has said openly that he works with current and former US Air Force and US Navy pilots, subject-matter experts, special-operations personnel and air traffic controllers to keep missions realistic without making them tedious.¹ Reflected Simulations, a frequent collaborator, has built campaigns including *Zone 5* for the F-14A; the two studios have jointly produced multi-aircraft campaigns — *Arctic Thunder* paired an

F/A-18C storyline from *Baltic Dragon* with F-16C missions from *Reflected Simulations* inside the same fictional Nordic scenario.² It is a genuinely small industry — a handful of studios, each with a recognisable authorial voice — closer to tabletop-RPG campaign writing than to conventional game-studio production.

This is also where the "hard by design" ethos from Chapter Three earns its keep. A *Baltic Dragon* or *Reflected Simulations* campaign assumes the buyer already owns and can operate the aircraft module in question — DCS: F-16C, DCS: F/A-18C — competently before the story missions begin; the campaigns are not tutorials, they are the payoff for having already climbed the module's own learning curve.

THE CHATTER — BUYING A CAMPAIGN AUTHOR, NOT JUST A PRODUCT

Community discussion of new campaign announcements reads closer to anticipating a favourite novelist's next book than a typical game-DLC launch: buyers reference specific authors' past work, debate mission-design choices against earlier campaigns by the same studio, and treat a *Baltic Dragon* release as a known quantity in a way most add-on purchases in this hobby are not.³

THE SCEPTIC

A small, name-driven market for scripted content is also a fragile one: DCS's entire single-player campaign ecosystem currently rests on the continued output of a handful of individuals and small studios, with no evidence of a broader pipeline behind them. If *Baltic Dragon* or *Reflected Simulations* slowed down, the campaign side of this hobby would visibly thin out — a concentration risk the community itself does not discuss often, but should.

2

THE NUMBER OF STUDIOS — BALTIC DRAGON AND REFLECTED SIMULATIONS — THIS ISSUE'S RESEARCH FOUND DOMINATING DCS WORLD'S SINGLE-PLAYER CAMPAIGN MARKET BY NAME RECOGNITION AND OUTPUT VOLUME.

Buying a DCS campaign reads less like buying downloadable content and more like buying the next book from an author whose last three you already own.

SOURCES ¹ Baltic Dragon campaign catalogue and SME collaboration process, baltic-dragon.net and Stormbirds.blog Q&A (9 May 2024). ² Reflected Simulations' DCS: F-14A *Zone 5* campaign, digitalcombatsimulator.com; *Arctic Thunder* joint campaign announcement, Stormbirds.blog (10 Aug 2023). ³ Community reception pattern for DCS campaign launches, Stormbirds.blog and DCS forum discussion, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER TEN · THE LOCAL ANGLE, HONESTLY SCOPED

A Real vACC, a Thin DIY Scene

Singapore has a genuine, named, active piece of this hobby's infrastructure — and no evidence of the home-cockpit-building meetup culture that exists in the UK or the US.

The strongest verifiable Singapore-specific piece of this hobby is SINvACC — the Singapore virtual Area Control Centre, a named, organised sub-unit of VATSIM's Southeast Asia division (VATSEA) that trains volunteer controllers to staff Singapore's actual Flight Information Region, centred on Changi (WSSS), using real charts and real procedures.¹ It sits inside VATSIM's wider Asia-Pacific structure — eight divisions running from Afghanistan to New Zealand — alongside sister vACCs for Malaysia, Thailand, the Philippines, Vietnam-Cambodia-Laos and Hong Kong.² A dedicated MSFS-focused Discord group for Singapore-based pilots also exists, organised around VATSIM flying and plane-spotting rather than hardware or DIY builds.³ This is real, active infrastructure, not a token mention — a Singapore-based enthusiast who wants to control real-feeling Changi traffic on VATSIM has an actual local organisation to join.

What this issue could not find is evidence of a Singapore-specific home-cockpit-building scene comparable to the UK or US examples in Chapter Five — no local meetup group, no simpit-builder forum thread, no equivalent to David Naylor's Barnsley front room. Commercial flight-experience venues exist and are genuinely worth naming: HAITE Aviation Safety and Training runs a Level-D-certified full-motion Airbus A320 simulator open to the public for 60- or 120-minute paid sessions, and WingsAcademy.sg markets itself as Singapore's largest flight-simulator centre.⁴ But both are commercial entertainment-and-training venues serving the general public and aspiring pilots, not gathering points for DIY hobbyists who own their own rigs — a distinction this issue is drawing deliberately rather than letting "flight simulator centre in Singapore" imply a hobbyist community that this research did not actually find.

THE SCEPTIC

It would be easy to pad this section by treating HAITE and WingsAcademy as evidence of a thriving local hobbyist scene. They are not that — they are businesses selling a flight-simulator experience to the public, which is a different thing from a community of enthusiasts who own home-cockpit hardware and gather to discuss it. The honest finding is a real, active VATSIM presence and a DIY scene this research could not verify exists in any organised form.⁵

THE CHATTER — WHAT A SINGAPORE-BASED ENTHUSIAST ACTUALLY JOINS

For a Singapore-based reader who wants a genuine local foothold in this hobby rather than a paid experience, the honest starting point is SINvACC itself — a real volunteer organisation with real training, not a storefront — followed by the general regional VATSEA and global Discords covered in Chapter Seven (p15), since no Singapore-specific hardware or simpit-building community surfaced in this research.

Singapore has a real vACC and no visible simpit-builder scene — which is not a contradiction, just an honest description of where this particular hobby currently sits in one small, aviation-obsessed city-state.

SOURCES 1 SINvACC (sinvacc.org), Singapore virtual Area Control Centre, VATSEA (VATSIM Southeast Asia) member division. 2 VATSIM Asia Pacific region structure (8 divisions), vat-apac.com. 3 "MSFS Singapore Pilots Discord Group," Microsoft Flight Simulator Forums community-groups listing. 4 HAITE Aviation Safety and Training, haite.com.sg (A320 Level D full-motion simulator, public 60/120-minute sessions); WingsAcademy.sg self-description as Singapore's largest flight-simulator centre. 5 Editorial assessment; no Singapore-specific home-cockpit-building community forum, meetup group or equivalent to Chapter Five's UK/US examples located in this issue's research.

CHAPTER TEN · THE JARGON, TRANSLATED

The Glossary — Every Term This Issue Used Without Stopping to Explain

Twenty terms, plainly defined, grouped the way the hobby itself divides them: the software, the hardware, and the radio.

8

VATSIM DIVISIONS COVERING THE ASIA-PACIFIC REGION ALONE, FROM AFGHANISTAN TO NEW ZEALAND — SINGAPORE'S SINVACC (P21) IS ONE SMALL, REAL PIECE OF A GENUINELY LARGE VOLUNTEER STRUCTURE.

SOFTWARE & SIM TERMS

HOTAS. "Hands On Throttle And Stick" — a joystick and throttle designed as one matched, integrated unit.¹

Study level. A contested community label for an add-on aircraft detailed enough to be studied, not just flown (Chapter Four, p9).

Cold and dark. An aircraft with all systems powered off, as it would sit overnight — the traditional, harder starting point for a "proper" startup.

Freeware / payware. Free versus paid add-ons; FlyByWire's A32NX (Chapter Four) is the hobby's best-known freeware example.

Blade element theory. X-Plane's method of calculating flight by modelling small segments of wings and props individually, rather than reading a pre-built performance table (Chapter Two, p6).

COMMUNITY & ATC TERMS

VATSIM / IVAO. The two major volunteer online air-traffic networks (Chapter Six, p13).

vACC. Virtual Area Control Centre — a VATSIM sub-organisation responsible for one region's virtual airspace, e.g. Singapore's SINvACC (p21).

FIR. Flight Information Region — the real-world airspace division a vACC virtually staffs.

Rating ladder (S1-C1). VATSIM's controller progression from student to full centre/enroute controller (Chapter Six, p14).

HARDWARE TERMS

Yoke. The wheel-and-column control used in most GA and airliner aircraft, as opposed to a fighter-style stick.

Throttle quadrant. The lever cluster controlling engine power, propeller pitch and mixture — modelled by hardware like Honeycomb's Bravo (Chapter Five, p11).

Force feedback (FFB). Motorised resistance in a stick or yoke that pushes back against the pilot's hand, rather than a passive spring.

Head tracking. Software (TrackIR and similar) that moves the in-sim camera to match the pilot's real head movement, without a full VR headset.

Simpit. A home-built physical cockpit shell, as opposed to a desk-mounted stick and monitor (Chapter Five, p12).

COMBAT-SIM TERMS

Module. A single purchasable aircraft in DCS World's otherwise free core platform (Chapter Eight, p17).

Campaign. A scripted single-player mission series, usually third-party, built for one specific DCS module (Chapter Nine, p20).

Presence. The VR-specific sensation of genuinely occupying a cockpit's space, distinct from simply seeing it (Chapter Nine, p19).

THE CHATTER — JARGON AS A MEMBERSHIP TEST

Newcomer threads across AVSIM and r/flightsim consistently ask for a glossary because the vocabulary itself functions as a soft membership test: knowing the difference between a module and an add-on, or between cold-and-dark and a running start, signals real time-in-hobby the way any subculture's jargon does.²

SOURCES 1 HOTAS, TrackIR, force-feedback and related hardware terminology, ffbeast.github.io and manufacturer documentation, cross-checked. 2 Newcomer glossary-request pattern, AVSIM/r/flightsim, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER ELEVEN · WHAT THE COMMUNITY ADMITS

The Upgrade Treadmill Has No Summit

Ten chapters of this issue have described a hobby with real depth. This one is about the parts of it that are, on the community's own admission, a trap.

Every tier of this hobby contains a version of the same honest confession, made in the community's own forums rather than in this issue's framing: the next purchase always promises to be the one that finally delivers "real" immersion, and it is very rarely the last one bought. Chapter Five's "buy once, cry once" hardware advice is sound in isolation and, in aggregate, a perfect mechanism for normalising an ever-higher baseline spend — today's high-end VIRPIL-and-VR setup is tomorrow's mid-tier expectation, not because the earlier gear stopped working, but because the comparison point moved.¹ DCS World's per-module pricing model (Chapter Eight, p17) produces the same pattern in miniature: each individual \$80 module is a defensible

purchase, and the accumulated module library after several years is, by the community's own recurring complaint, expensive in a way no single purchase ever felt like it would be.

Diminishing returns are real and measurable in at least one dimension this issue can point to directly: the jump from a \$50 stick to a \$1,200 mid-tier setup changes the actual flying experience enormously; the jump from a \$1,200 mid-tier setup to a \$4,000 high-end VR rig changes it far less, in proportion to the money spent, than the marketing around premium hardware implies. Nobody in this hobby's own community disputes that pattern when asked directly — the disagreement is only ever about where, exactly, the curve bends.

THE SCEPTIC, IN FULL

This issue's own Chapter Two told the cautionary story from the industry side: Microsoft's 2012 *Flight* reboot collapsed within months after a hostile reception and no add-on ecosystem to redeem it — proof that even the biggest name in the hobby's history can misjudge what this community actually values.² The consumer-side mirror of that failure is quieter and has no single collapse to point to: it is the individual enthusiast who spends the equivalent of a decent used car on a simpit build that never quite gets finished, or who owns four premium HOTAS units purchased in successive "this is definitely the last upgrade" moments. Neither story is unique to flight simming — every serious hobby has this shape — but this one's per-item price points (a single DCS module at \$80, a single study-level airliner at \$70–90, a single VIRPIL throttle north of \$300) make the treadmill unusually easy to step onto one small, individually reasonable purchase at a time.³

\$0

WHAT IT COSTS TO FIND OUT WHETHER THIS HOBBY IS FOR YOU AT ALL — VATSIM, IVAO, MSFS'S OWN FREE TRIAL TIERS AND FLYBYWIRE'S A32NX ARE ALL FREE. THE TREADMILL STARTS ONLY AFTER THAT.

The disagreement in this hobby's own forums is never about whether the treadmill exists. It is only ever about where, exactly, a given enthusiast's personal exit ramp is supposed to be.

THE CHATTER — NAMING IT OUT LOUD

Threads titled some variant of "am I spending too much on this" recur across every forum this issue researched, and the replies follow a predictable, almost ritual shape: gentle reassurance, a joke about a spouse's reaction to a delivery box, and — buried in most of them — someone admitting they have, in fact, spent more than they are comfortable saying out loud.⁴

SOURCES 1 "Buy once, cry once" hardware-advice pattern and its aggregate effect, synthesised from Chapters Five and Eight sourcing and community hardware-forum discussion, Aug 2026. 2 *Microsoft Flight* (2012) reboot and withdrawal, per Chapter Two sourcing (note 5, p5). 3 Per-item pricing (DCS module \$80, study-level airliner \$70–90, VIRPIL throttle \$300+) per Chapters Four, Five and Eight sourcing. 4 "Am I spending too much" thread pattern, AVSIM/r/flightsim/Hoggit, Aug 2026 research — characterised, not quoted verbatim.

CHAPTER ELEVEN · A WAY IN

An Honest Ninety Days, With Real Prices

A starter path built entirely from products and prices already verified in this issue — spend nothing until day thirty, and know exactly what the next sixty are for.

The single best piece of advice this research surfaced, repeated across every forum covered in Chapter Seven, is to spend the first month finding out which of the three communities in Chapter One actually holds an enthusiast's interest before buying anything beyond the absolute minimum. Free tiers exist specifically to make this possible: Microsoft Flight Simulator and X-Plane both offer free or heavily discounted trial access, DCS World's core platform and Caucasus map are free outright, and FlyByWire's A32NX (Chapter Four) is a genuinely excellent freeware airliner, not a demo.¹ A keyboard-

and-mouse or a basic controller is sufficient for this entire first month.

Only after that does a first real purchase make sense — and it should be shaped by which of the three worlds actually held attention, not by which YouTube video looked most impressive. This issue's own reporting supports one general rule: buy the cheapest hardware in the tier one level above where an enthusiast expects to stay, because the community consensus across every forum surveyed is that people under-buy their first controller and over-buy their fourth one.

DAYS 1–30 — SPEND NOTHING

Install MSFS or X-Plane's free/trial tier, or DCS World's free core with the Caucasus map.

Fly FlyByWire's free A32NX if the civil/airliner side appeals — it is not a lesser product for being free.

Read, don't buy: AVSIM's forums, r/flightsim, and — if combat sim appeals — the Hoggit Discord.

Create a VATSIM account and fly one observed flight into controlled airspace, even with no hardware beyond a keyboard.

DAYS 31–90 — ONE REAL PURCHASE

Civil/GA path: a Thrustmaster T-Flight Hotas X (~\$75) plus one study-level aircraft that matches a genuine interest (~\$70–90).

Combat path: DCS World's free core plus exactly one module matching the aircraft that hooked the enthusiast in Days 1–30 (~\$40–80).

Either path: start a VATSIM pilot rating alongside the hardware — it costs nothing and pays off regardless of platform.

Do not buy rudder pedals, a VR headset, or a second aircraft/module in this window. That decision belongs to month four, made with real hours logged.

Ninety days, one real purchase, and a free ATC rating in progress — everything after that is optional, and the community's own history says to treat it that way.

THE CHATTER — WHAT VETERANS WISH THEY'D KNOWN

Asked what they would tell their own first-month selves, veteran enthusiasts converge on a short, consistent list: try before buying whenever a free tier exists, resist a full simpit build before finishing a single real flight lesson's worth of hours, and treat the first hardware purchase as a two-year commitment, not a two-week one — advice this issue's Ninety Days path was built to follow, not just repeat.²

\$0–165

THE REALISTIC ALL-IN COST OF THIS ISSUE'S NINETY DAYS PATH — FREE THROUGH DAY 30, ONE HARDWARE-PLUS-CONTENT PURCHASE AFTER THAT, NOTHING ELSE.

SOURCES ¹ Free/trial access: Microsoft Flight Simulator and X-Plane trial tiers, DCS World free core platform, FlyByWire A32NX (GPLv3) — per Chapters Two, Four and Six sourcing. Starter-path pricing per Chapters Four and Five. ² Veteran-advice-to-newcomers pattern, AVSIM/r/flightsim, Aug 2026 research — characterised, not quoted verbatim.

THE LONG READ

The Cockpit That Isn't There

A century before Microsoft's servers buckled under simulator demand, a man who could not afford flying lessons built one out of organ parts — and the affliction Wilbur Wright confessed to in 1900 has been finding cheaper ways to indulge itself ever since.

In 1927, a young man named Edwin Link left school wanting badly to learn to fly and could not afford the lessons. Rather than give up the ambition, he spent eighteen months building an alternative: a stubby wooden fuselage on a universal joint, pitching and rolling on organ bellows and compressed air borrowed from his father's piano-and-organ factory in Binghamton, New York.¹ He introduced it on 14 April 1929. It looked, by every contemporary account, like an overgrown toy. It was not treated as one for long: over the next two decades, and especially through the Second World War, more than half a million airmen from nearly every combatant nation learned to fly, at least in part, inside a machine that never left the ground.² The affliction Wilbur Wright confessed to Octave Chanute in the spring of 1900 — "for some years I have been afflicted with the belief that flight is possible to man" (p2) — did not require the machine to be real. It required only that the belief be indulged, by whatever means were affordable.

Read against that history, the eleven chapters preceding this one describe the same affliction finding its late-2020s price points. A photoreal render of an actual neighbourhood loading outside a simulated cockpit window (p3) is doing, cheaply and instantly, what Link's bellows did slowly and mechanically: standing in for a real sky that money, geography or nerve put out of reach. VATSIM's roughly 190,000 active volunteers (p13) are doing, for free and for no institutional mandate, what military flight schools paid instructors to do with Link Trainers — building a shared, rule-governed airspace that behaves like the real one because enough people agreed to make it behave that way. And DCS World's verified use by real A-10 pilots training in VR (p19) is the closest thing this issue found to Link's own trajectory: a machine built for enthusiasts, taken seriously enough by the institution it imitated to become, briefly, genuinely useful to it.

Link built his machine because the real sky was too expensive. Ninety-seven years later, the real sky is still too expensive for most of the people in this issue — it is simply a different kind of expensive now, measured in Honeycomb throttles and VIRPIL sticks instead of flight-school tuition.

500,000+

AIRMEN TRAINED, AT LEAST IN PART, ON LINK'S ORGAN-BELLOWS MACHINE DURING THE SECOND WORLD WAR — A HOBBYIST'S COMPENSATION THAT BECAME INDISPENSABLE WARTIME INFRASTRUCTURE.

THE MECHANISM

Link's machine worked as training infrastructure for a specific reason worth naming plainly: it was cheap and repeatable in exactly the ways a real aircraft, at wartime scale, could not be. A cadet who crashed a Link Trainer walked away and tried again in minutes; a cadet who crashed a real aircraft did not. That asymmetry — failure made free, then made instructive — is the same mechanism Chapter Four's study-level aircraft rely on when they model a checklist error as a real, consequence-bearing failure rather than a game-over screen (p9). The machine changes; the reason it works has not.

SOURCES ¹ Edwin Link, Link Trainer origin (1927–29 development, Binghamton, NY, organ-factory components), Vintage Aviation News and ASME International engineering-history landmark documentation. ² Link Trainer WWII training scale (over 500,000 airmen, "almost every combatant nation"), per the same sourcing, cross-checked against Naval Air Station Fort Lauderdale Museum's Link Trainer history.

THE LONG READ, CONTINUED

A Century-Old Word for the Same Feeling

100+ yrs

HOW LONG AMATEUR ASTRONOMERS HAVE CALLED THEIR OWN UPGRADE COMPULSION "APERTURE FEVER" — THE SAME AFFLICTION, A DIFFERENT SKY, A CENTURY OF PRIOR ART.

There is an older precedent, too, for Chapter Eleven's upgrade treadmill (p23) — and it comes with its own century-old name. Amateur astronomers have called the compulsion to buy an ever-larger telescope "aperture fever" for more than a hundred years; the phenomenon reportedly met its first practical ceiling only with the construction of the 36-inch Lick and 40-inch Yerkes refractors early in the twentieth century, when the physics of grinding a bigger lens simply ran out of road.¹ The word choice is not incidental. Astronomers call it a fever; Wright called his own compulsion an affliction. Both hobbies, a century apart and pointed at opposite directions in the sky, reached for the language of illness to describe the same experience: a desire that each purchase promises to satisfy and does not, because the goalpost is the comparison to whatever the next enthusiast owns, not any fixed technical need.

What makes this issue's specific version of the fever worth taking seriously — rather than simply filing it

alongside every other gear-acquisition hobby — is how much unpaid, skilled labour it has produced along the way. Baltic Dragon and Reflected Simulations (p20) consult working relationships with actual military pilots to write DCS campaigns nobody has to buy. FlyByWire's contributors built and gave away an Airbus simulation good enough that payware developers are implicitly measured against it (p9). SINvACC's volunteers (p21) learn real separation standards to control virtual traffic over a real airport, for an audience that could simply fly unsupervised instead. None of that labour was necessary. All of it happened anyway, for the same reason Edwin Link spent eighteen months hand-building a machine nobody had asked him to build: because the alternative to a cheap, partial version of the thing was no version of the thing at all, and that turned out to be intolerable to enough people to build an entire hobby's infrastructure around avoiding it.

Nobody required Baltic Dragon to consult real fighter pilots for a campaign nobody has to buy, or SINvACC's volunteers to learn real separation standards for traffic that could simply fly unsupervised. They did it anyway — which is either the most reassuring or the most alarming fact in this entire issue, depending on how contagious an affliction has to be before "hobby" stops being the right word for it.

THE THREADS

Five things this issue kept finding in the same shape: (1) simulated flight as a cheaper substitute for a real one, going back at least to 1929; (2) volunteer labour building real-feeling infrastructure — VATSIM's controllers, DCS's campaign writers — for no pay; (3) a systems-modelling culture that treats faithful failure as more valuable than easy success; (4) a genuine, if narrow, crossover into real-world training that this issue could verify in at least one documented case; and (5) an upgrade treadmill with a name in at least two unrelated hobbies, a century apart, both reaching for the language of sickness to describe it.²

SOURCES 1 "Aperture fever," amateur-astronomy terminology and historical context (Lick/Yerkes refractor era), BBC Sky at Night Magazine and Cloudy Nights community discussion, cross-checked. 2 Synthesis of Chapters One through Eleven of this issue; see each chapter's own sourcing for the underlying facts.

THE LAST WORD

A reader flagged an article in chat and asked for the full treatment. Eleven chapters, one long history and a great many verified prices later, the honest summary is this: it is a real hobby, with a real community, real volunteer infrastructure, and at least one documented case of the real US Air Force training on it. It is also, on its own community's admission, a treadmill with no marked summit. Both things are true at once, and this issue has tried to say so plainly rather than pick the flattering half.

SOURCES, IN BRIEF — flightsimulator.com, x-plane.com, digitalcombatsimulator.com, prepar3d.com, lockheedmartin.com (the four simulators); PMDG, Fenix Simulations, iniBuilds and flybywiresim.com (study-level and freeware aircraft); flyhoneycomb.com, winwingsim.com, virpil-controls.eu/us.com, mfg-sim.com (hardware); vatsim.net, sinvacc.org, vat-apac.com, ivao.aero, pilotedge.net (online ATC networks); avsim.com, wiki.hoggitworld.com, discord.com/invite/hoggit, flightsimexpo.com (community and events); baltic-dragon.net and Stormbirds.blog (DCS campaigns); haite.com.sg and wingsacademy.sg (Singapore flight-experience venues); the War Zone/twz.com (A-10 VR training); Library of Congress (Wilbur Wright correspondence); vintageaviationnews.com and asme.org (the Link Trainer); BBC Sky at Night Magazine (aperture fever). Full per-page footnotes throughout. Figures and prices as researched and dated Aug 2026; single-source claims flagged in place; nothing in this issue was invented.